# Type MS · Explain how smoking damages health, including in pregnancy

## Answer 1 of 6 · Question 2 part d · 4 marks

2022 JAN · Paper 02 · extract 1 of 2 · find part d in this mark scheme extract

| Number | | | | |
|---|---|---|---|---|
| 2 | (a) (i) | an answer that makes reference to three of the following:<br>• warms air;<br>• moistens air;<br>• layer of (sticky) mucus;<br>• removes/traps, pathogens/bacteria/microorganisms/dust / dirt (from entering lungs);<br>• reference to 1st line of defence; | Mp3 must be linked to mp4 for mark | 3 |
| | (ii) | • prevents collapse;<br>• to allow air to flow freely (to lungs/alveoli);<br>• allows neck to bend; | | 2 |
| | (iii) | B; (bronchi) | | 1 |
| | (b) | D; (medulla oblongata) | | 1 |
| | (c) | $17 \times 0.50 = 8.5\ dm^3$;<br>$4 \times 8.5$;<br>$= 0.34\ dm^3$; | | 3 |
| | (d) | • reduced oxygen uptake by haemoglobin/forms carboxyhaemoglobin;<br>• less oxygen to developing fetus;<br>• lower rate of respiration/less energy released (by fetus);<br>• less growth/lower birth mass; | | 4 |

Total 14 marks

# Type MS · Explain how smoking damages health, including in pregnancy

## Answer 1 of 6 · Question 2 part d · 4 marks

2022 JAN · Paper 02 · extract 2 of 2 · find part d in this mark scheme extract

| Question number | Answer | Notes | Marks |
|---|---|---|---|

Type MS · Explain how smoking damages health, including in pregnancy

Answer 2 of 6 · Question 3 part c · 5 marks

2019 JUN · Paper 02 (regional R) · extract 1 of 2 · find part c in this mark scheme extract

| | | | |
|---|---|---|---|
| **3(a)(i)** | B 3.2 - 3.5 kg; | | **(1)** |

| Question Number | Answer | Additional guidance | Mark |
|---|---|---|---|
| **3(a)(ii)** | (94 / 210) x 100 ;<br>45%; (2) | Ecf allow correct answer to incorrect figures used in calculation<br><br>Allow 44.76%/ 44.8% for correct final answer | **(2)** |

| Question Number | Answer | Additional guidance | Mark |
|---|---|---|---|
| **3(b)** | smoking appears to lower birth mass;<br>more higher birth mass babies born to non-smokers; | ORA for both mp's | **(2)** |

| **Question Number** | Answer | Additional guidance | Mark |
|---|---|---|---|
| **3(c)** | • cigarette smoke contains CO;<br>• combines with haemoglobin instead of oxygen/forms carboxyhaemoglobin;<br>• less oxygen to fetus;<br>• less respiration in fetus;<br>• lower growth rate/cell division;<br>• causes lower birth mass; | | **(5 max)** |

**Total 10 Marks**

## Type MS · Explain how smoking damages health, including in pregnancy

Answer 2 of 6 · Question 3 part c · 5 marks

2019 JUN · Paper 02 (regional R) · extract 2 of 2 · find part c in this mark scheme extract

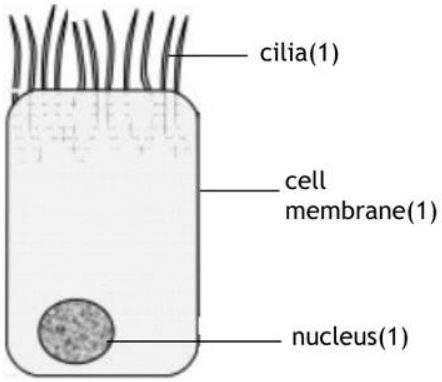
**Paper 2: Subject Human Biology**

| Question Number | Answer | Additional guidance | Mark |
|---|---|---|---|

Type MS · Explain how smoking damages health, including in pregnancy

Answer 3 of 6 · Question 2 part b.i · 1 mark

2020 NOV · Paper 01 (regional R) · find part b.i in this mark scheme extract

| | | | |
|---|---|---|---|
| 2 (a) (i) |  | | 3 |
| (ii) | cilia → holds the genetic code<br>cell membrane → controls entry and exit of<br>nucleus → site where most chemical reactions occur<br>makes proteins<br>moves mucus through the trachea | Reject more than one line from each structure | 3 |
| (b) (i) | Tar (1) | | 1 |
| (ii) | • (cilia)burned/destroyed/reduced in number(1)<br>• paralysed/cannot beat to and fro/move mucus(1) | | 2 |

**Total for Question 2 = 9 Marks**

## Type MS · Explain how smoking damages health, including in pregnancy

### Answer 4 of 6 · Question 3 part a · 2 marks

2024 JUN · Paper 02 · find part a in this mark scheme extract

| Question number | | | | |
|---|---|---|---|---|
| 3 | (a) | Smoking cigarettes;<br>Causes (lung) cancer/emphysema/bronchitis/heart disease/eq ;<br><br>OR<br><br>lack of exercise;<br>obesity/heart disease/high blood pressure/ type ii diabetes/eq; | Allow drugs or named drug and their effect on health for 2 marks. | (2) |
| | (b) | Any two from:<br>• production of bile reduced/affected;<br>• less fat emulsified/fat not emulsified;<br>• lipase less active/lipase unable to digest as much (fat)/slower lipase activity; | Allow bile not produced<br><br>Allow description of emulsification | (2) |
| | (c) | blurred/misty vision/blindness/less light passing through the lens / less light reaching the retina; | | (1) |
| | (d) | cerebrum/cerebellum/ cerebral cortex/cerebral hemispheres/hypothalamus/medulla (oblongata); | Allow any part of the brain that is affected by alcohol | (1) |
| | (e) | Any three from:<br>• reduces release of neurotransmitters;<br>• reduced diffusion across synapse;<br>• less neurotransmitters bind to receptors on post-synaptic membrane;<br>• slows/ fewer transmission of impulses;<br>• slower reaction /reflexes; | | (3) |
| | (f) | Any two from<br>• increased heart rate;<br>• increased breathing rate;<br>• pupil dilation;<br>• increase in blood sugar/glucose levels/ increased breakdown of glycogen to glucose;<br>• increase in blood flow to brain/muscles;<br>• increase in (aerobic) respiration (in muscles); | | (2) |

**Total for Question 3 = 11 marks**

Type MS · Explain how smoking damages health, including in pregnancy

Answer 5 of 6 · Question 4 part a.iii · 2 marks

2025 JUN · Paper 02 (regional R) · find part a.iii in this mark scheme extract

| | | | | | |
|---|---|---|---|---|---|
| 4 | (a) | (i) | • decreases;<br>• from 82% to 72%/by 10% | | 2 |
| | | (ii) | 79:21;<br>3.76:1; | accept 3.7612:1<br>accept 3.762:1<br>accept 3.8/4:1 | 2 |
| | | (iii) | any two from<br>• passive smoking/ex- smokers already have damaged lungs;<br>• (air) pollution;<br>• genetics/heredity; | | 2 |
| | | (iv) | tar; | | 1 |
| | (b) | (i) | bronchiole; | | 1 |
| | | (ii) | alveolus; | | 1 |
| | | (iii) | any four from<br>• damaged alveoli/alveoli walls broken;<br>• less surface area (for gas exchange);<br>• difficulty breathing/shortness of breath;<br>• less oxygen passing into the blood/ less carbon dioxide removed from blood/less gaseous exchange;<br>• reduced (aerobic) respiration;<br>• fatigue/less energy/not very active; | | 4 |
| | | | | **Total for Question 4 = 13 marks** | |

Type MS · Explain how smoking damages health, including in pregnancy

Answer 6 of 6 · Question 4 part a.iv · 1 mark

2025 JUN · Paper 02 (regional R) · find part a.iv in this mark scheme extract

| | | | | | |
|---|---|---|---|---|---|
| 4 | (a) | (i) | • decreases;<br>• from 82% to 72%/by 10% | | 2 |
| | | (ii) | 79:21;<br>3.76:1; | accept 3.7612:1<br>accept 3.762:1<br>accept 3.8/4:1 | 2 |
| | | (iii) | any two from<br>• passive smoking/ex- smokers already have damaged lungs;<br>• (air) pollution;<br>• genetics/heredity; | | 2 |
| | | (iv) | tar; | | 1 |
| | (b) | (i) | bronchiole; | | 1 |
| | | (ii) | alveolus; | | 1 |
| | | (iii) | any four from<br>• damaged alveoli/alveoli walls broken;<br>• less surface area (for gas exchange);<br>• difficulty breathing/shortness of breath;<br>• less oxygen passing into the blood/ less carbon dioxide removed from blood/less gaseous exchange;<br>• reduced (aerobic) respiration;<br>• fatigue/less energy/not very active; | | 4 |
| | | | | **Total for Question 4 = 13 marks** | |